

MATH245 (Honours Algebra I) Notes

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Table of Contents

I	Some Language and Notation of Mathematics	2
1	Sets	2
1.1	Basic Set Theory	2
1.2	Basic Set Operations	3
1.3	Algebra of Set Operations	5

Part I

Some Language and Notation of Mathematics

1 Sets

1.1 Basic Set Theory

Definition 1.1.1

A **set** is a collection of elements.

Intuitively, a set can be thought of as a bag—our set—which may or may not contain stuff—our elements. Sets are the most fundamental objects in mathematics and many important concepts, such as functions and spaces, can be defined using sets.

If we want to say that an element x belongs to some set S , we write

$$x \in S$$

with the “ $\in$ ” symbol meaning “an element of” (so $x \in S$ literally means x is an element of S).

When defining a set, we use curly braces “ $\{ \}$ ” to enclose the elements inside of it.

Example 1.1.2. The following are some basic examples of sets:

1. $A := \{0, 3, \heartsuit\}$ defines a set, named A , to the elements 0, 3, and $\heartsuit$ (elements do not necessarily need to be numbers, indeed another set can also be an element!)
2. $B := \{\text{all odd integers}\}$ defines a set, named B , to all odd integers (e.g. $-1 \in B$ and $127 \in B$).
3. $C := \{n : n \text{ is an integer satisfying } n^3 = n\}$ (the colon “ $:$ ” in between the curly braces stands for “such that,” so C is the set of all integers n such that $n^3 = n$ (equivalently, $C := \{-1, 0, 1\}$))
4. The standard sets:

$\mathbb{N} := \{0^1, 1, 2, 3, \dots\}$ (the set of natural numbers)

$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}$ (the set of integers)

$\mathbb{Q} := \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$ (the set of rational numbers)

$\mathbb{R} :=$ the set of real numbers²

$\mathbb{C} := \{a + bi : a, b \in \mathbb{R}, i = \sqrt{-1}\}$ (the set of complex numbers)

¹Some sources omit 0 from the set of natural numbers, and define $\mathbb{N} := \{1, 2, 3, \dots\}$. However in this class, we will always assume $0 \in \mathbb{N}$, and define $\mathbb{N}^+ := \{1, 2, 3, \dots\}$.

²The set of real numbers is surprisingly difficult to define. Take MATH255 to learn how to rigorously define $\mathbb{R}$.

Remark 1.1.3. Order and repetition in sets do not matter. Indeed, the set $A := \{0, 3, \heartsuit\}$ is equal to the sets $A_1 := \{3, \heartsuit, 0\}$ and $A_2 := \{0, 3, \heartsuit, 0\}$.

Definition 1.1.4

Two sets A and B are said to be equal if and only if they have the same elements. Symbolically, we write $A = B \iff \forall x, x \in A \text{ when } x \in B$.

In the above theorem, we introduced two key symbols often used in mathematics. The first, the “ $\iff$ ” symbol, stands for “if and only if.” In mathematics, this means that the statement on the left is equivalent to the statement on the right. The second symbol, the “ $\forall$ ”, stands for “for all” or “for every” and is called the **universal quantifier**. Symbolic notation is often eschewed in formal writing but is used very often in more casual mathematical writing.

Remark 1.1.5. By Definition 1.0.3, we see that the set A in Remark 1.0.2 is equal to the sets A_1 and A_2 .

Definition 1.1.6

Let A and B be arbitrary sets. We say that A is a **subset of** B , or A is **contained in** B , if for every $x \in A$, $x \in B$. In this case we write, $A \subseteq B$, where “ $\subseteq$ ” is the subset symbol³.

Remark 1.1.7. Note that $A = B$ holds when $A \subseteq B$ and $B \subseteq A$ both hold. This gives us a very useful way to prove if two sets are equal (we simply prove that one set is the subset of the other and vice versa).

Definition 1.1.8

The set with no elements is called the **empty set**, and is denoted $\emptyset := \{\}$.

Intuitively, the empty set can be thought of as a bag with nothing inside of it. The empty set is *not* nothing, but it contains nothing.

Example 1.1.9. We conclude with some examples of sets and some of their properties:

1. For every set A , $\emptyset \subseteq A$.
2. $\emptyset \neq \{\emptyset\}$, indeed we see that $\emptyset \subseteq \{\emptyset\}$ but $\{\emptyset\} \not\subseteq \emptyset$.
3. $\{\emptyset, \{\emptyset\}\} \neq \{\{\emptyset\}\}$
4. In set theory, we write the numbers $0 := \emptyset$, $1 := \{\emptyset\}$, $2 := \{0, 1\}$, and so on.

1.2 Basic Set Operations

Now that we have a basic understanding of sets, we will now learn how to work with multiple sets. What do we do if we want to compare elements across different sets? For instance, how can

³Many textbooks use the symbol “ $\subset$ ” to denote a subset instead of $\subseteq$, i.e. if A is a subset of B , then $A \subset B$ is written. In these notes, I will use the symbol $\subset$ only when one set is a *proper subset* of the other, in the same way that $<$ defines a strict inequality.

we “add” and “subtract” elements of different sets, or find elements that they have in common? This subsection aims to introduce the fundamental operations on sets that help us answer these questions.

Definition 1.2.10

Let A and B be two sets. Define:

- Their **union**, denoted $A \cup B$, as the set $\{x: x \in A \text{ or } x \in B\}$
- Their **intersection**, denoted $A \cap B$, as the set $\{x: x \in A \text{ and } x \in B\}$
- Their **difference**, denoted $A \setminus B$, as the set $\{x: x \in A \text{ and } x \notin B\}$.

If A is an ambient set (i.e. there are other sets that are contained in A), then if $B \subseteq A$ we say the **complement** of B relative to A , denoted B^c , is the difference:

$$B^c := A \setminus B$$

Example 1.2.11. Within $\mathbb{Z}$, define $A := \{n \in \mathbb{Z}: n \text{ is even}\}$. We have that the complement of A relative to $\mathbb{Z}$, $A^c := \{n \in \mathbb{Z}: n \text{ is odd}\}$.

Definition 1.2.12

For two elements a and b (a and b are not necessarily sets), we define their **ordered pair**, denoted (a, b) , to be the tuple satisfying the property:

$$(a, b) = (c, d) \quad \text{if and only if} \quad a = c \text{ and } b = d$$

In particular, if $a \neq b$, then $(a, b) \neq (b, a)$. $(a, b) \neq (b, a)$ whenever $a \neq b$.

We will use this notion of ordered pairs to help us with the following important definition:

Definition 1.2.13

Let A, B be two sets. We define their **Cartesian product**, denoted $A \times B$, as the set of ordered pairs $\{(a, b): a \in A \text{ and } b \in B\}$.

A notable Cartesian product is the Euclidean plane $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$. You have already seen this plane defined as the xy -plane, where each ordered pair (x, y) represents the x - and y -coordinates of a point on the plane.

Example 1.2.14. Let $A := \{0, 1, 2\}$ and $B := \{a, b\}$. Then their Cartesian product $A \times B$ is the set $\{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\}$.

Remark 1.2.15. In general, if A has n elements and B has m elements, then their cartesian product $A \times B$ has $n \cdot m$ elements.

Definition 1.2.16

Let A be a set. We say the **power set of A** , denoted 2^A or $\mathcal{P}(A)$, is the set of all subsets of A , which includes A and the empty set $\emptyset$.

Example 1.2.17. Let $A := \{0, 1, 2\}$, then its power set $\mathcal{P}(A)$ is:

$$\mathcal{P}(A) := \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$$

Remark 1.2.18. For each element in a nonempty set, we can either choose to include or not include it in a subset of A . This means that for a set A with n elements, its power set has 2^n elements.

1.3 Algebra of Set Operations

We will show our first proposition (a statement which we will prove) here.

Proposition 1.3.19

Let A , B , and C be arbitrary sets. Then the following properties hold:

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
3. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
4. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$
5. Given an ambient set U such that $A \subseteq U$, $(A^c)^c = A$

Proof. Each statement in this proposition asserts that two sets are equal. By Remark 1.0.6, we prove each equality by showing that each side is a subset of the other.

Let us start by proving (1). We first show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$. Let $x \in A \cap (B \cup C)$. By definition, $x \in A$ and $x \in B \cup C$. Therefore, $x \in B$ or $x \in C$.

If $x \in B$, then $x \in A \cap B$, and hence $x \in (A \cap B) \cup (A \cap C)$. If $x \in C$, then $x \in A \cap C$, and hence $x \in (A \cap B) \cup (A \cap C)$. Therefore,

$$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C).$$

Conversely, let $x \in (A \cap B) \cup (A \cap C)$. Then $x \in A \cap B$ or $x \in A \cap C$.

If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $x \in A \cap (B \cup C)$. If $x \in A \cap C$, then $x \in A$ and $x \in C$, so $x \in A \cap (B \cup C)$. Therefore,

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C).$$

Hence condition (1) holds.

For (2), let $x \in A \cup (B \cap C)$. If $x \in A$, then $x \in A \cup B$ and $x \in A \cup C$. Therefore, $x \in (A \cup B) \cap (A \cup C)$.

If $x \in B \cap C$, then $x \in B$ and $x \in C$. Thus $x \in A \cup B$ and $x \in A \cup C$, so again $x \in (A \cup B) \cap (A \cup C)$. Hence

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C).$$

Conversely, let $x \in (A \cup B) \cap (A \cup C)$. Then $x \in A \cup B$ and $x \in A \cup C$.

If $x \in A$, then $x \in A \cup (B \cap C)$. Otherwise, $x \notin A$. Since $x \in A \cup B$ and $x \notin A$, we must have $x \in B$. Similarly, since $x \in A \cup C$ and $x \notin A$, we must have $x \in C$. Thus $x \in B \cap C$, and therefore $x \in A \cup (B \cap C)$. Hence

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C),$$

so condition (2) holds.

For (3), let $x \in A \setminus (B \cup C)$. By definition, $x \in A$ and $x \notin B \cup C$. Since $x \notin B \cup C$, we have $x \notin B$ and $x \notin C$; otherwise, x would belong to $B \cup C$. Thus $x \in A \setminus B$ and $x \in A \setminus C$. Therefore,

$$x \in (A \setminus B) \cap (A \setminus C),$$

and hence

$$A \setminus (B \cup C) \subseteq (A \setminus B) \cap (A \setminus C).$$

Conversely, let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A \setminus B$ and $x \in A \setminus C$, which means $x \in A$, $x \notin B$, and $x \notin C$. Since $x \notin B$ and $x \notin C$, it follows that $x \notin B \cup C$. Thus $x \in A \setminus (B \cup C)$. Therefore,

$$(A \setminus B) \cap (A \setminus C) \subseteq A \setminus (B \cup C),$$

and condition (3) holds.

For (4), let $x \in A \setminus (B \cap C)$. By definition, $x \in A$ and $x \notin B \cap C$. Therefore, $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in A \setminus B$. If $x \notin C$, then $x \in A \setminus C$. Hence $x \in (A \setminus B) \cup (A \setminus C)$. Therefore,

$$A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C).$$

Conversely, let $x \in (A \setminus B) \cup (A \setminus C)$. Then $x \in A \setminus B$ or $x \in A \setminus C$. Thus $x \in A$ and either $x \notin B$ or $x \notin C$. Therefore, $x \notin B \cap C$, so $x \in A \setminus (B \cap C)$. Hence

$$(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C),$$

and condition (4) holds.

Finally, by definition of the complement relative to U ,

$$A^c = U \setminus A.$$

Therefore,

$$(A^c)^c = U \setminus (U \setminus A) = A.$$

Hence condition (5) holds. ■

Proposition 1.3.20 (De Morgan's Laws)

Let U be a set, and let $B, C \subseteq U$. Then:

1. $(B \cap C)^c = B^c \cup C^c$.
2. $(B \cup C)^c = B^c \cap C^c$.

Proof. For (1), let $x \in (B \cap C)^c$. By definition of the complement, $x \in U$ and $x \notin B \cap C$. Therefore, it is not the case that both $x \in B$ and $x \in C$. Hence $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in B^c$. If $x \notin C$, then $x \in C^c$. In either case, $x \in B^c \cup C^c$. Thus

$$(B \cap C)^c \subseteq B^c \cup C^c.$$

Conversely, let $x \in B^c \cup C^c$. Then $x \in B^c$ or $x \in C^c$, so $x \notin B$ or $x \notin C$. Therefore, $x \notin B \cap C$. Since $x \in U$, it follows that $x \in (B \cap C)^c$. Hence

$$B^c \cup C^c \subseteq (B \cap C)^c.$$

Therefore,

$$(B \cap C)^c = B^c \cup C^c.$$

For (2), let $x \in (B \cup C)^c$. By definition, $x \in U$ and $x \notin B \cup C$. Therefore, $x \notin B$ and $x \notin C$. Thus $x \in B^c$ and $x \in C^c$, which implies that

$$x \in B^c \cap C^c.$$

Hence

$$(B \cup C)^c \subseteq B^c \cap C^c.$$

Conversely, let $x \in B^c \cap C^c$. Then $x \in B^c$ and $x \in C^c$, meaning that $x \notin B$ and $x \notin C$. Therefore, $x \notin B \cup C$. Since $x \in U$, we conclude that $x \in (B \cup C)^c$. Thus

$$B^c \cap C^c \subseteq (B \cup C)^c.$$

Therefore,

$$(B \cup C)^c = B^c \cap C^c.$$

■